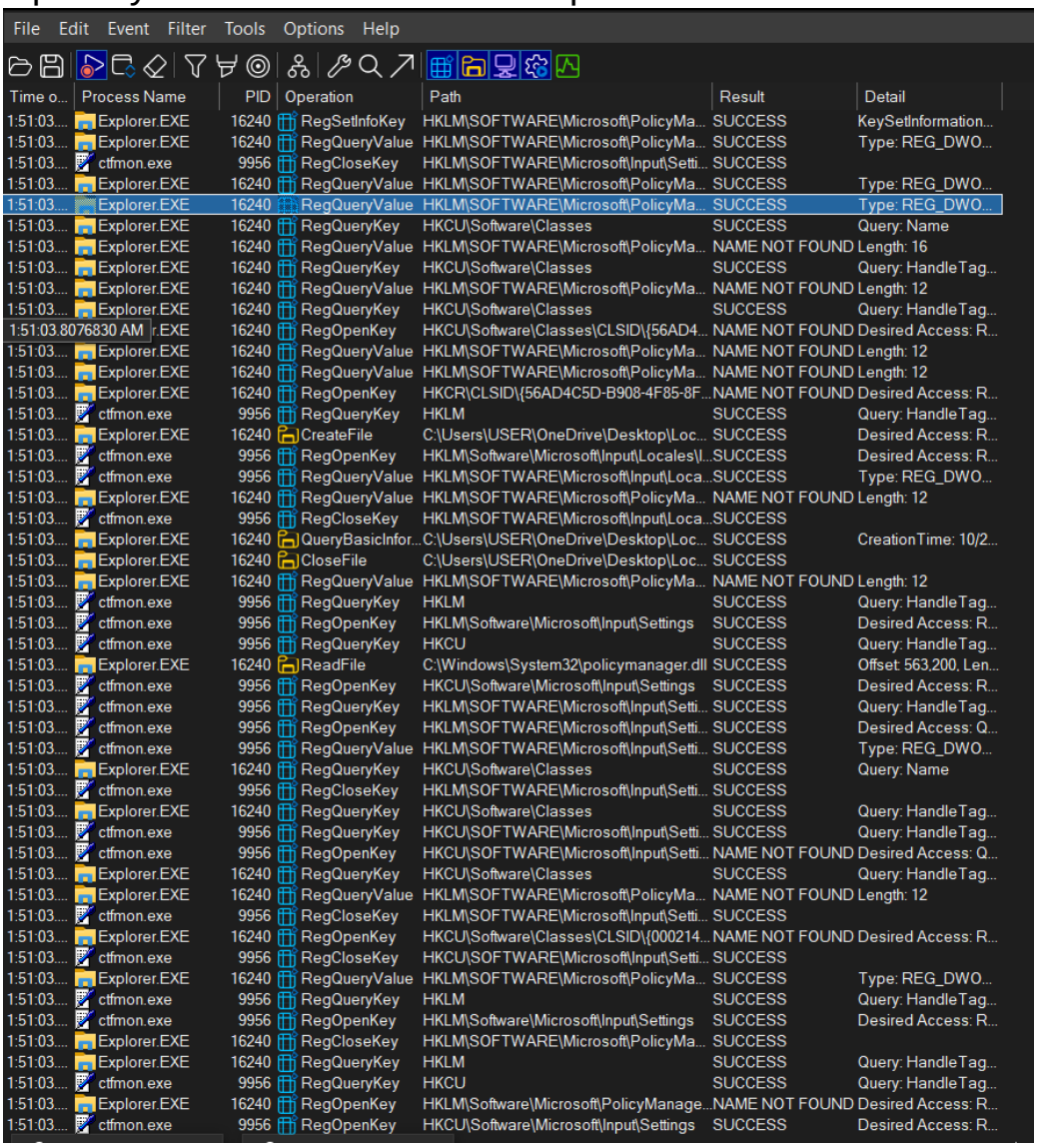
Windows Internals Deep Dive: Analyzing ctfmon.exe & Explorer.exe with Sysinternals.

Scope and Objectives:

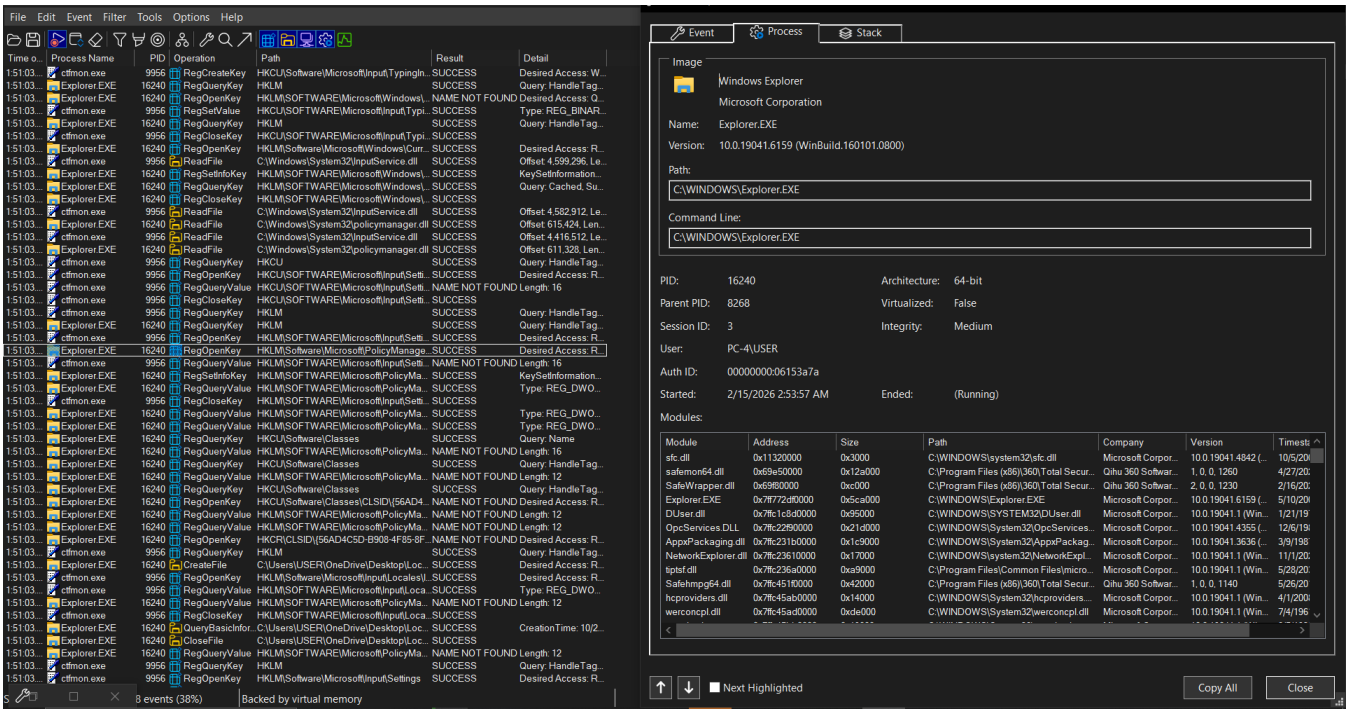
- Observe and characterize registry, file, and process activity
- Validate the authenticity and integrity of core system binaries
- Identify indicators of compromise (IOCs), anomalous behavior, or policy violations
- Establish whether activity aligns with standard Windows behavior

1. Open Sysinternals and click on Explorer.EXE with PID **16240**.



Time o...	Process Name	PID	Operation	Path	Result	Detail
1:51:03....	Explorer.EXE	16240	RegSetInfoKey	HKLM\SOFTWARE\Microsoft\PolicyMa...	SUCCESS	KeySetInformation...
1:51:03....	Explorer.EXE	16240	RegQueryValue	HKLM\SOFTWARE\Microsoft\PolicyMa...	SUCCESS	Type: REG_DWO...
1:51:03....	ctfmon.exe	9956	RegCloseKey	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	
1:51:03....	Explorer.EXE	16240	RegQueryValue	HKLM\SOFTWARE\Microsoft\PolicyMa...	SUCCESS	Type: REG_DWO...
1:51:03....	Explorer.EXE	16240	RegQueryValue	HKLM\SOFTWARE\Microsoft\PolicyMa...	SUCCESS	Type: REG_DWO...
1:51:03....	Explorer.EXE	16240	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: Name
1:51:03....	Explorer.EXE	16240	RegQueryValue	HKLM\SOFTWARE\Microsoft\PolicyMa...	NAME NOT FOUND	Length: 16
1:51:03....	Explorer.EXE	16240	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: HandleTag...
1:51:03....	Explorer.EXE	16240	RegQueryValue	HKLM\SOFTWARE\Microsoft\PolicyMa...	NAME NOT FOUND	Length: 12
1:51:03....	Explorer.EXE	16240	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: HandleTag...
1:51:03.8076830 AM	r.EXE	16240	RegOpenKey	HKCU\Software\Classes\CLSID\{56AD4...	NAME NOT FOUND	Desired Access: R...
1:51:03....	Explorer.EXE	16240	RegQueryValue	HKLM\SOFTWARE\Microsoft\PolicyMa...	NAME NOT FOUND	Length: 12
1:51:03....	Explorer.EXE	16240	RegQueryValue	HKLM\SOFTWARE\Microsoft\PolicyMa...	NAME NOT FOUND	Length: 12
1:51:03....	Explorer.EXE	16240	RegOpenKey	HKCR\CLSID\{56AD4C5D-B908-4F85-8F...	NAME NOT FOUND	Desired Access: R...
1:51:03....	ctfmon.exe	9956	RegQueryKey	HKLM	SUCCESS	Query: HandleTag...
1:51:03....	Explorer.EXE	16240	CreateFile	C:\Users\USER\OneDrive\Desktop\Loc...	SUCCESS	Desired Access: R...
1:51:03....	ctfmon.exe	9956	RegOpenKey	HKLM\Software\Microsoft\Input\Locales\I...	SUCCESS	Desired Access: R...
1:51:03....	ctfmon.exe	9956	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Loca...	SUCCESS	Type: REG_DWO...
1:51:03....	Explorer.EXE	16240	RegQueryValue	HKLM\SOFTWARE\Microsoft\PolicyMa...	NAME NOT FOUND	Length: 12
1:51:03....	ctfmon.exe	9956	RegCloseKey	HKLM\SOFTWARE\Microsoft\Input\Loca...	SUCCESS	
1:51:03....	Explorer.EXE	16240	QueryBasicInfor...	C:\Users\USER\OneDrive\Desktop\Loc...	SUCCESS	CreationTime: 10/2...
1:51:03....	Explorer.EXE	16240	CloseFile	C:\Users\USER\OneDrive\Desktop\Loc...	SUCCESS	
1:51:03....	Explorer.EXE	16240	RegQueryValue	HKLM\SOFTWARE\Microsoft\PolicyMa...	NAME NOT FOUND	Length: 12
1:51:03....	ctfmon.exe	9956	RegQueryKey	HKLM	SUCCESS	Query: HandleTag...
1:51:03....	ctfmon.exe	9956	RegOpenKey	HKLM\Software\Microsoft\Input\Settings	SUCCESS	Desired Access: R...
1:51:03....	ctfmon.exe	9956	RegQueryKey	HKCU	SUCCESS	Query: HandleTag...
1:51:03....	Explorer.EXE	16240	ReadFile	C:\Windows\System32\policymanager.dll	SUCCESS	Offset: 563,200, Len...
1:51:03....	ctfmon.exe	9956	RegOpenKey	HKCU\Software\Microsoft\Input\Settings	SUCCESS	Desired Access: R...
1:51:03....	ctfmon.exe	9956	RegQueryKey	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	Query: HandleTag...
1:51:03....	ctfmon.exe	9956	RegOpenKey	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	Desired Access: Q...
1:51:03....	ctfmon.exe	9956	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	Type: REG_DWO...
1:51:03....	Explorer.EXE	16240	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: Name
1:51:03....	ctfmon.exe	9956	RegCloseKey	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	
1:51:03....	Explorer.EXE	16240	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: HandleTag...
1:51:03....	ctfmon.exe	9956	RegQueryKey	HKCU\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	Query: HandleTag...
1:51:03....	ctfmon.exe	9956	RegOpenKey	HKCU\SOFTWARE\Microsoft\Input\Setti...	NAME NOT FOUND	Desired Access: Q...
1:51:03....	Explorer.EXE	16240	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: HandleTag...
1:51:03....	Explorer.EXE	16240	RegQueryValue	HKLM\SOFTWARE\Microsoft\PolicyMa...	NAME NOT FOUND	Length: 12
1:51:03....	ctfmon.exe	9956	RegCloseKey	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	
1:51:03....	Explorer.EXE	16240	RegOpenKey	HKCU\Software\Classes\CLSID\{000214...	NAME NOT FOUND	Desired Access: R...
1:51:03....	ctfmon.exe	9956	RegCloseKey	HKCU\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	
1:51:03....	Explorer.EXE	16240	RegQueryValue	HKLM\SOFTWARE\Microsoft\PolicyMa...	SUCCESS	Type: REG_DWO...
1:51:03....	ctfmon.exe	9956	RegQueryKey	HKLM	SUCCESS	Query: HandleTag...
1:51:03....	ctfmon.exe	9956	RegOpenKey	HKLM\Software\Microsoft\Input\Settings	SUCCESS	Desired Access: R...
1:51:03....	Explorer.EXE	16240	RegCloseKey	HKLM\SOFTWARE\Microsoft\PolicyMa...	SUCCESS	
1:51:03....	Explorer.EXE	16240	RegQueryKey	HKLM	SUCCESS	Query: HandleTag...
1:51:03....	ctfmon.exe	9956	RegQueryKey	HKCU	SUCCESS	Query: HandleTag...
1:51:03....	Explorer.EXE	16240	RegOpenKey	HKLM\Software\Microsoft\PolicyManage...	NAME NOT FOUND	Desired Access: R...
1:51:03....	ctfmon.exe	9956	RegOpenKey	HKCU\Software\Microsoft\Input\Settings	SUCCESS	Desired Access: R...

2. Click on process for full analysis for the Explorer.EXE with PID **16240**, PPID **8268**



Explorer.exe (Windows Explorer)

- **Executable Path:** `C:\Windows\Explorer.exe`
- **Integrity Level:** Medium
- **Signature:** Microsoft Corporation
- **Function:** Provides the Windows shell (desktop, taskbar, file browsing)

Behavioral Observation:
Explorer.exe executed extensive registry reads targeting user profile classes, shell extensions, and Group Policy values. Loaded DLLs included both Microsoft-signed components and legitimate third-party security modules (e.g., endpoint protection hooks). No unsigned or suspicious libraries were present.

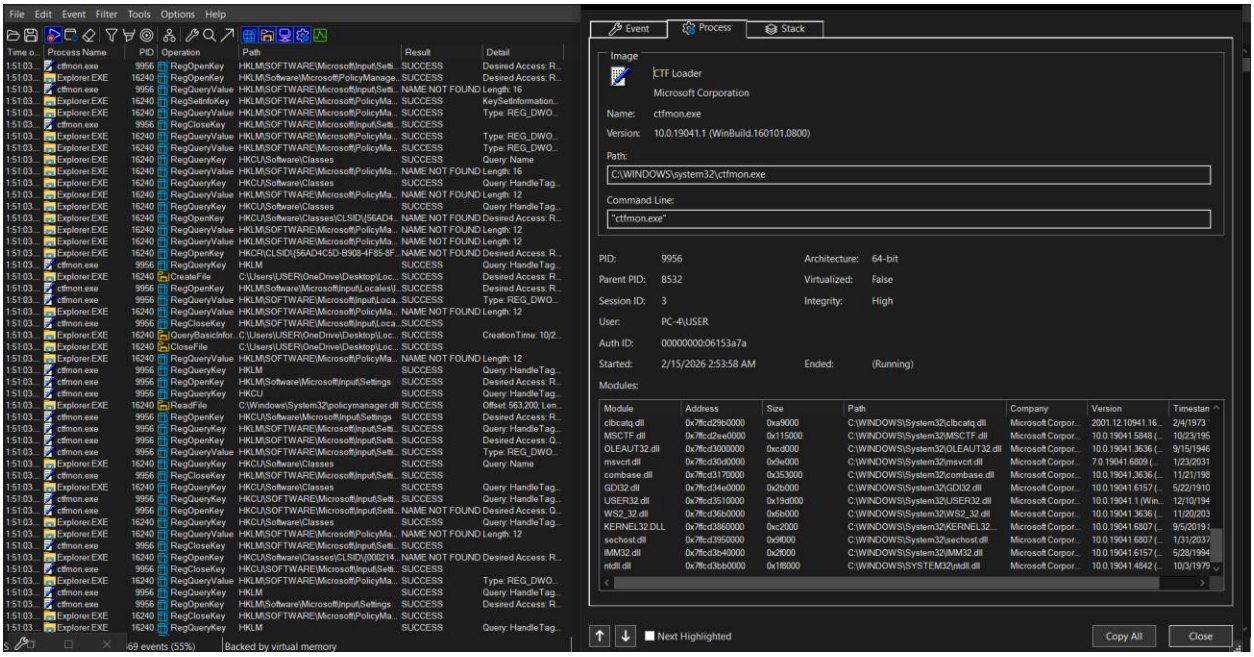
Assessment:
Observed behavior aligns with expected shell initialization. No signs of malicious code injection or unauthorized persistence.

Taking a closer look at process metadata, paths, signatures, integrity levels, and loaded modules and taking a look at the PID and the PPID confirmed that the Explorer.EXE process was behaving exactly as expected. All binaries were Microsoft-signed, registry calls were clean, and no anomalous DLL injections or suspicious persistence mechanisms were present.

3. Click on the ctfmon.exe with the PID **9956**

Time o...	Process Name	PID	Operation	Path	Result	Detail
1:51:03....	Explorer.EXE	16240	RegSetInfoKey	HKLM\SOFTWARE\Microsoft\PolicyMa...	SUCCESS	KeySetInformation...
1:51:03....	Explorer.EXE	16240	RegQueryValue	HKLM\SOFTWARE\Microsoft\PolicyMa...	SUCCESS	Type: REG_DWO...
1:51:03....	ctfmon.exe	9956	RegCloseKey	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	
1:51:03....	Explorer.EXE	16240	RegQueryValue	HKLM\SOFTWARE\Microsoft\PolicyMa...	SUCCESS	Type: REG_DWO...
1:51:03....	Explorer.EXE	16240	RegQueryValue	HKLM\SOFTWARE\Microsoft\PolicyMa...	SUCCESS	Type: REG_DWO...
1:51:03....	Explorer.EXE	16240	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: Name
1:51:03....	Explorer.EXE	16240	RegQueryValue	HKLM\SOFTWARE\Microsoft\PolicyMa...	NAME NOT FOUND	Length: 16
1:51:03....	Explorer.EXE	16240	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: HandleTag...
1:51:03....	Explorer.EXE	16240	RegQueryValue	HKLM\SOFTWARE\Microsoft\PolicyMa...	NAME NOT FOUND	Length: 12
1:51:03....	Explorer.EXE	16240	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: HandleTag...
1:51:03....	Explorer.EXE	16240	RegOpenKey	HKCU\Software\Classes\CLSID\{56AD4...	NAME NOT FOUND	Desired Access: R...
1:51:03....	Explorer.EXE	16240	RegQueryValue	HKLM\SOFTWARE\Microsoft\PolicyMa...	NAME NOT FOUND	Length: 12
1:51:03....	Explorer.EXE	16240	RegQueryValue	HKLM\SOFTWARE\Microsoft\PolicyMa...	NAME NOT FOUND	Length: 12
1:51:03....	Explorer.EXE	16240	RegOpenKey	HKCR\CLSID\{56AD4C5D-B908-4F85-8F...	NAME NOT FOUND	Desired Access: R...
1:51:03....	ctfmon.exe	9956	RegQueryKey	HKLM	SUCCESS	Query: HandleTag...
1:51:03....	Explorer.EXE	16240	CreateFile	C:\Users\USER\OneDrive\Desktop\Loc...	SUCCESS	Desired Access: R...
1:51:03....	ctfmon.exe	9956	RegOpenKey	HKLM\Software\Microsoft\Input\Locales\l...	SUCCESS	Desired Access: R...
1:51:03....	ctfmon.exe	9956	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Loca...	SUCCESS	Type: REG_DWO...
1:51:03....	Explorer.EXE	16240	RegQueryValue	HKLM\SOFTWARE\Microsoft\PolicyMa...	NAME NOT FOUND	Length: 12
1:51:03....	ctfmon.exe	9956	RegCloseKey	HKLM\SOFTWARE\Microsoft\Input\Loca...	SUCCESS	
1:51:03....	Explorer.EXE	16240	QueryBasicInfor...	C:\Users\USER\OneDrive\Desktop\Loc...	SUCCESS	CreationTime: 10/2...
1:51:03....	Explorer.EXE	16240	CloseFile	C:\Users\USER\OneDrive\Desktop\Loc...	SUCCESS	
1:51:03....	Explorer.EXE	16240	RegQueryValue	HKLM\SOFTWARE\Microsoft\PolicyMa...	NAME NOT FOUND	Length: 12
1:51:03....	ctfmon.exe	9956	RegQueryKey	HKLM	SUCCESS	Query: HandleTag...
1:51:03....	ctfmon.exe	9956	RegOpenKey	HKLM\Software\Microsoft\Input\Settings	SUCCESS	Desired Access: R...
1:51:03....	ctfmon.exe	9956	RegQueryKey	HKCU	SUCCESS	Query: HandleTag...
1:51:03....	Explorer.EXE	16240	ReadFile	C:\Windows\System32\policymanager.dll	SUCCESS	Offset: 563,200, Len...
1:51:03....	ctfmon.exe	9956	RegOpenKey	HKCU\Software\Microsoft\Input\Settings	SUCCESS	Desired Access: R...
1:51:03....	ctfmon.exe	9956	RegQueryKey	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	Query: HandleTag...
1:51:03....	ctfmon.exe	9956	RegOpenKey	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	Desired Access: Q...
1:51:03....	ctfmon.exe	9956	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	Type: REG_DWO...
1:51:03....	Explorer.EXE	16240	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: Name
1:51:03....	ctfmon.exe	9956	RegCloseKey	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	
1:51:03....	Explorer.EXE	16240	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: HandleTag...
1:51:03....	ctfmon.exe	9956	RegQueryKey	HKCU\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	Query: HandleTag...
1:51:03....	ctfmon.exe	9956	RegOpenKey	HKCU\SOFTWARE\Microsoft\Input\Setti...	NAME NOT FOUND	Desired Access: Q...
1:51:03....	Explorer.EXE	16240	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: HandleTag...
1:51:03....	Explorer.EXE	16240	RegQueryValue	HKLM\SOFTWARE\Microsoft\PolicyMa...	NAME NOT FOUND	Length: 12
1:51:03....	ctfmon.exe	9956	RegCloseKey	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	
1:51:03....	Explorer.EXE	16240	RegOpenKey	HKCU\Software\Classes\CLSID\{000214...	NAME NOT FOUND	Desired Access: R...
1:51:03....	ctfmon.exe	9956	RegCloseKey	HKCU\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	
1:51:03....	Explorer.EXE	16240	RegQueryValue	HKLM\SOFTWARE\Microsoft\PolicyMa...	SUCCESS	Type: REG_DWO...
1:51:03....	ctfmon.exe	9956	RegQueryKey	HKLM	SUCCESS	Query: HandleTag...
1:51:03....	ctfmon.exe	9956	RegOpenKey	HKLM\Software\Microsoft\Input\Settings	SUCCESS	Desired Access: R...
1:51:03....	Explorer.EXE	16240	RegCloseKey	HKLM\SOFTWARE\Microsoft\PolicyMa...	SUCCESS	
1:51:03....	Explorer.EXE	16240	RegQueryKey	HKLM	SUCCESS	Query: HandleTag...
1:51:03....	ctfmon.exe	9956	RegQueryKey	HKCU	SUCCESS	Query: HandleTag...
1:51:03....	Explorer.EXE	16240	RegOpenKey	HKLM\Software\Microsoft\PolicyManage...	NAME NOT FOUND	Desired Access: R...
1:51:03....	ctfmon.exe	9956	RegOpenKey	HKCU\Software\Microsoft\Input\Settings	SUCCESS	Desired Access: R...

4. Click on process for the analysis of ctfmon.exe with the PID **9956** and PPID **8532**.



ctfmon.exe (CTF Loader)

- **Executable Path:** `C:\Windows\System32\ctfmon.exe`
- **Signature:** Microsoft Corporation
- **Integrity Level:** High
- **Function:** Manages text input services and the Text Services Framework

Behavioral Observation:
ctfmon.exe performed frequent ***RegOpenKey*** and ***RegQueryValue*** operations across Input settings, IMM configuration, and policy discovery keys. All operations succeeded and were read-only in nature. Loaded modules originated strictly from the Windows system directory.

Assessment:
Activity corresponds to normal initialization of Windows input services. No anomalies identified.

Taking a closer look at process metadata, paths, signatures, integrity levels, and loaded modules and taking a look at the PID and the PPID confirmed that the cftmon.exe process was behaving exactly as expected. All binaries were Microsoft-signed, registry calls were clean, and no anomalous DLL injections or suspicious persistence mechanisms were present.

Recommendations

Although no threat activity was found, the following best practices are advised:

- Maintain active monitoring of core system processes using Sysinternals and EDR platforms
- Periodically validate signature integrity of critical executables
- Establish a behavioral baseline for workstation environments to improve anomaly detection
- Retain logs for long-term forensic comparison and incident-readiness